PRACTICAL NO 4. BUSINESS INTELLIGENCE TOOLS FOR DATA ANALYSIS

Aim:- To utilize Tableau as a Business Intelligence tool to extract insights from a dataset and create interactive visualizations for effective data analysis.

Main Purpose:- The main purpose of this practical is to learn how **Tableau converts raw data into meaningful charts, graphs, and dashboards** so that business information can be understood easily and decisions can be made from it.

Dataset:-

You can use a simple **Retail Store Dataset**.

Order ID	Date	Product	Category	Region	Sales	Quantity	Profit
1001	01-01-2025	Laptop	Electronics	Mumbai	55000	2	8000
1002	03-01-2025	Mobile	Electronics	Pune	30000	3	4500
1003	05-01-2025	Chair	Furniture	Mumbai	12000	4	2500
1004	08-01-2025	Table	Furniture	Nashik	18000	2	3500
1005	10-01-2025	Printer	Electronics	Pune	15000	2	2200
1006	12-01-2025	Desk	Furniture	Mumbai	22000	3	4000
1007	15-01-2025	Keyboard	Accessories	Nashik	5000	5	1200
1008	18-01-2025	Mouse	Accessories	Pune	3000	6	800
1009	20-01-2025	Monitor	Electronics	Mumbai	25000	2	3500
1010	23-01-2025	Sofa	Furniture	Pune	35000	1	6000

Step-by-Step Procedure in Tableau

Step 1: Open Tableau

1. Open **Tableau Public** or **Tableau Desktop**.
2. Click **Connect to Data**.
3. Select **Microsoft Excel** if your dataset is in Excel.
4. Select your Excel file.
5. Select the required sheet.
6. Click **Sheet 1** at the bottom.

Step 2: Understand the Data

After opening the worksheet, you will see fields such as:

Dimensions

- Order ID
- Date
- Product
- Category
- Region

Measures

- Sales
- Quantity
- Profit

Simple meaning:-

Dimension = information used to categorize data.

Example:

Product, Category, Region

Measure = numerical information that can be calculated.

Example:

Sales, Quantity, Profit

Step 3: Create a Sales by Category Chart

We want to answer:

> **Which category has the highest sales?**

1. Open **Sheet 1**.
2. Drag **Category → Columns**.
3. Drag **Sales → Rows**.
4. Tableau automatically creates a **bar chart**.
5. Click **Show Me** if necessary and select **Bar Chart**.
6. Click **Sort Descending** to arrange categories from highest to lowest sales.

You will get a chart showing:

Category → Sales

Sales by Category

Illustrative example showing how category sales can be compared in a BI visualization.

035K70K105K140K
Electronics
Furniture
Accessories

Step 4: Create Sales by Region

Now we want to know:

> **Which region generates the highest sales?**

1. Create a new worksheet.
2. Drag **Region → Columns**.
3. Drag **Sales → Rows**.
4. Select **Bar Chart**.
5. Sort the bars.

This allows us to compare:

Mumbai vs Pune vs Nashik

Step 5: Create Profit by Category

We want to find:

> **Which category generates the highest profit?**

1. Create a new worksheet.
2. Drag **Category → Columns**.
3. Drag **Profit → Rows**.
4. Select **Bar Chart**.
5. Turn on **Show Mark Labels**.

Now the profit value will appear on each bar.

Step 6: Create Sales Trend

We can analyze how sales change over time.

1. Create a new worksheet.
2. Drag **Date → Columns**.
3. Drag **Sales → Rows**.
4. Tableau will create a **line chart**.
5. If necessary, change Date to **Day** or **Month**.

This helps us answer:

> **How are sales changing over time?**

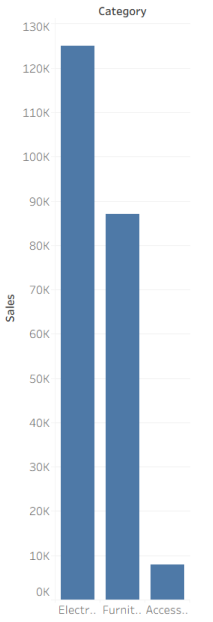
Step 7: Create a Sales by Category Pie Chart

We can also show the contribution of each category.

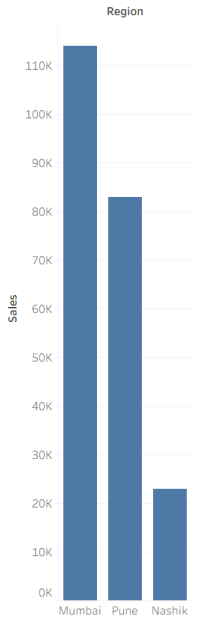
1. Create a new worksheet.
2. Drag **Category → Color**.
3. Drag **Sales → Angle**.
4. Change **Marks** from Automatic to **Pie**.
5. Drag **Sales → Label**.

This shows the proportion of total sales contributed by each category.

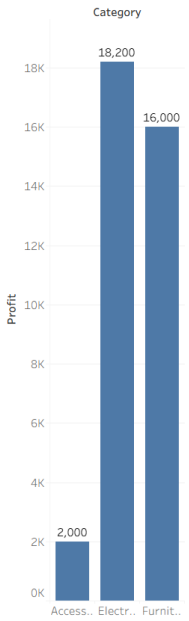
Sheet 1



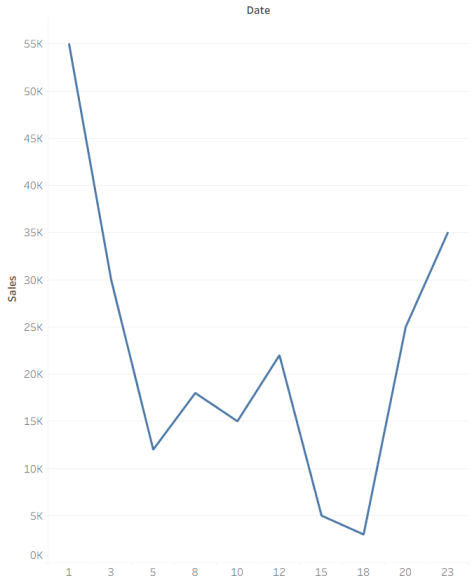
Sales by Region



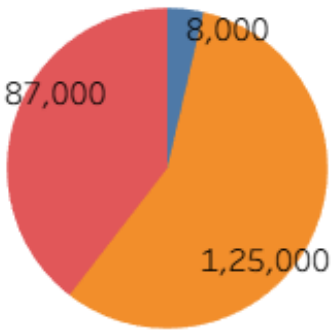
Profit by Category



Sales Trend



Sales By Category Pie Chart



Step 8: Add a Filter

This is an important part of **interactive visualization**.

For example, we can allow the user to select a **Region**.

1. Select your chart.
2. Drag **Region** to the **Filters** shelf.
3. Select **All**.
4. Click the filter drop-down.
5. Select **Show Filter**.

Now the user can select:

- Mumbai
- Pune
- Nashik

The chart will change according to the selected region.



Step 9: Create a Dashboard

Now we combine all the charts.

1. Click **New Dashboard** at the bottom.
2. Give it a name:

RETAIL STORE PERFORMANCE DASHBOARD

3. Drag the following sheets into the dashboard:
    - Sales by Category
    - Sales by Region
    - Profit by Category
    - Sales Trend
    - Sales by Category Pie Chart

Arrange them properly.

Step 10: Make the Dashboard Interactive

Add filters such as:

Region

Category

Date

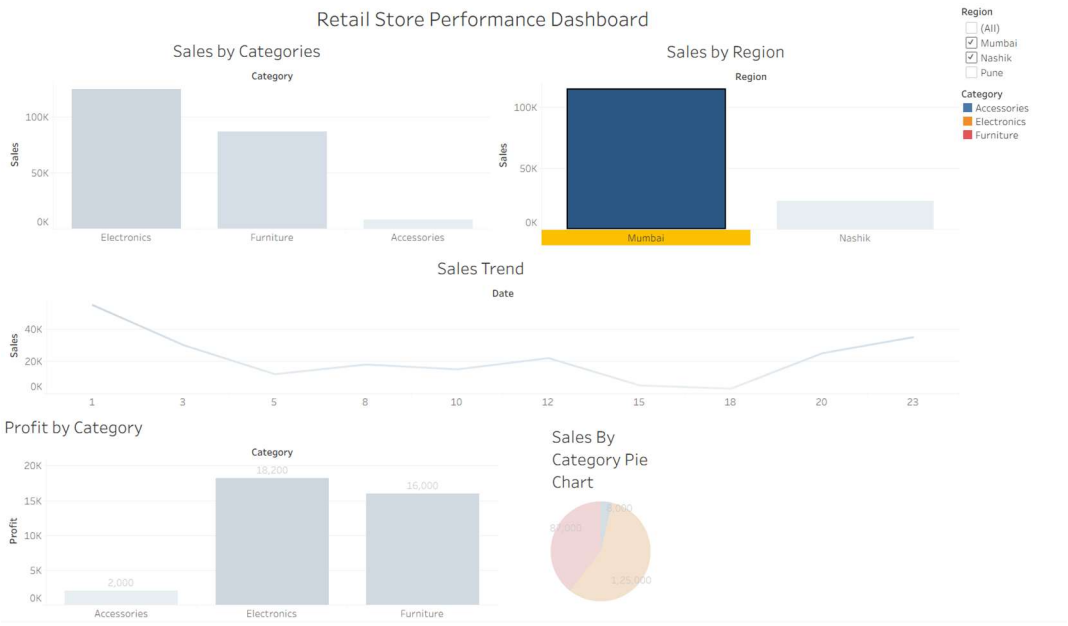
For example:

If the user selects:

Region = Mumbai

all relevant charts can update to show Mumbai's data.

This is what makes the dashboard **interactive**.



Step 11: Extract Business Insights

Finally, observe the charts and write meaningful conclusions.

For example:

Possible Insights

1. **Electronics** contributes the highest sales among the categories.
2. **Mumbai** generates higher sales compared with some other regions.
3. Furniture products contribute significant profit.
4. Sales vary across different dates.
5. The dashboard allows users to compare regions and categories interactively.

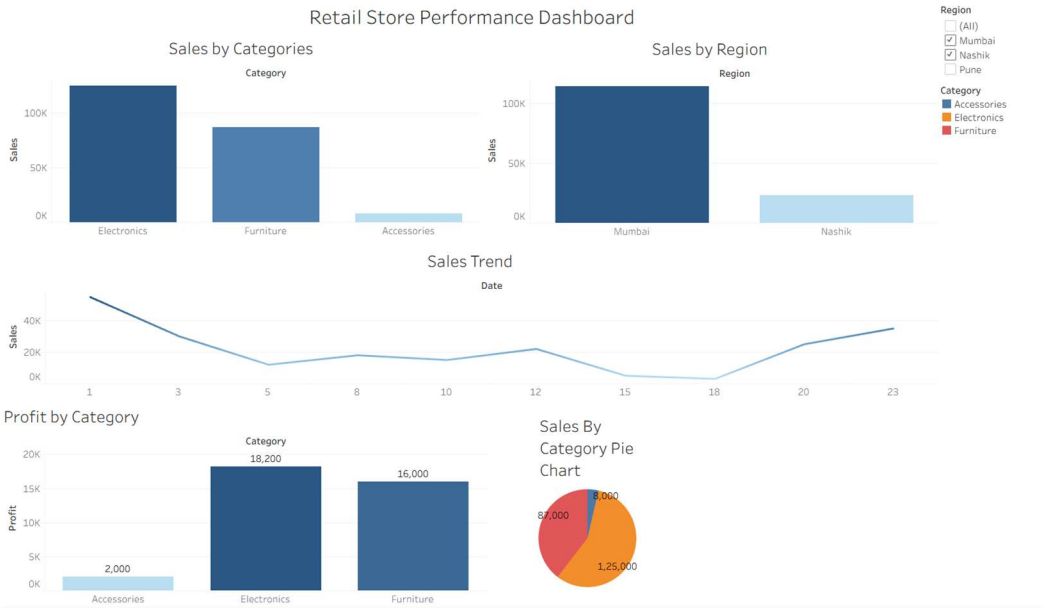
The exact insights should be written according to the results produced by your Tableau dataset.

Expected Output:-

You should finally have a Tableau dashboard containing:

RETAIL STORE PERFORMANCE DASHBOARD

- 📊 Sales by Category
- 📊 Sales by Region
- 📊 Profit by Category
- 📈 Sales Trend
- 🥧 Sales Contribution by Category
- 🔽 Region/Category filters



Practical Flow:-

Raw Dataset → Tableau → Data Analysis → Charts → Filters → Interactive Dashboard → Business Insights

Conclusion:-

Thus, this practical helped us understand how Tableau can be used as an effective Business Intelligence tool to analyze data and generate meaningful insights. We learned how to connect a dataset to Tableau, organize and analyze the data, and represent information using different charts and graphs, such as bar charts, line charts, and pie charts. We also understood how to apply filters and create interactive visualizations to explore data from different perspectives. These visualizations make complex information easier to understand and help identify trends, patterns, comparisons, and variations in business performance. Overall, this practical demonstrated how Tableau transforms raw data into useful business insights, supports data-driven decision-making, and improves the efficiency of data analysis and reporting.